CLIMATE SENSITIVE SHIPPING CONTAINER IN FOUR DIFFERENT CLIMATES PROJECT

1.0 INTRODUCTION

Worldwide, accommodation is a huge challenge and there should be creative solutions and approaches. The shipping container project is termed 'climate sensitive' and that simply means each should respond to the climate within which it resides. The radical container movement which is very popular amongst young people in recent times is worth noting as a design solution. After the ten-year active life of a shipping container, it would be useful to adapt these to address very challenging global issues such as housing. The project is supposed to be self sufficient as much as possible in terms of lighting, power, heating and cooling. The passive first approach is the best step in creating an architectural solutions that are climate sensitive. The project should consider the benefits that the microclimate of the site offers. The target user for this project is a recent university graduate who might reside anywhere around the globe. The container measures are; 12metres long x 2.4metres wide x 2.4 metres high

AIM

The aim of this project was to ascertain the knowledge of learners of how buildings thrive in different climates with a focus on passive design strategies, construction and materials.

Site Selection: Four distinct sites were chosen for the project.

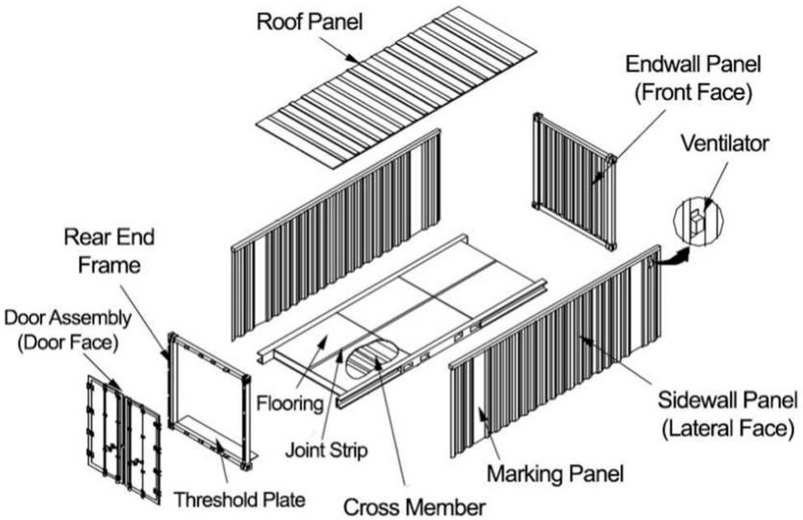
i. Cold climate – Whitehorse, Canada.
ii. Temperate climate – Clayton, United States of America.
iii. Warm and Humid climate – Accra, Ghana.
iv. Hot and Dry climate – Aswan, Egypt.

1.2 METHODS AND MODE OF DELIVERY

Project was realised through the following:

i. Online site survey of the chosen sites using google maps and google earth
ii. Detailed climate characteristics with climate consultant 6.0 app
iii. Pencil sketches and conceptual development
iv. Revit Architecture 2024 models
v. Rendered photorealistic models (Lumion 2024)
vi. Final poster designs with Canva and Ms PowerPoint

Shipping Container Anatomy



maps and google

COLD CLIMATE - Canada



South –East View In Summer



South –East View In Winter

TEMPERATE CLIMATE – United States of America



South –West View



North –West View

WARM HUMID CLIMATE - Ghana



North – East View



North – West View

HOT DRY CLIMATE- Egypt



North View



North-west View

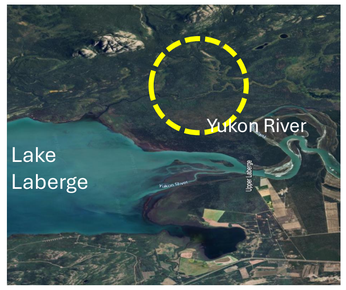
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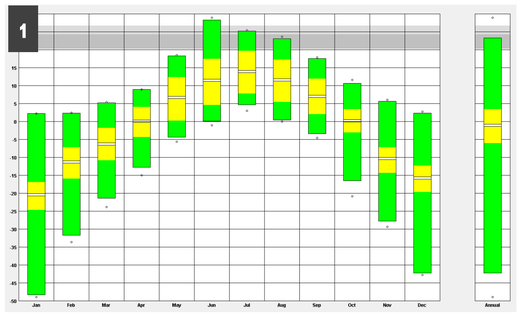
CLIMATE SENSITIVE SHIPPING CONTAINER - Cold Climate

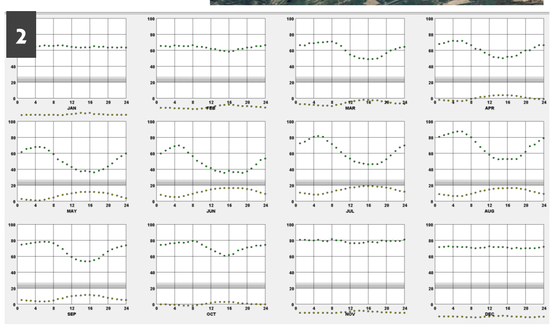
LOCATION : Whitehorse, along Yukon river, CANADA.

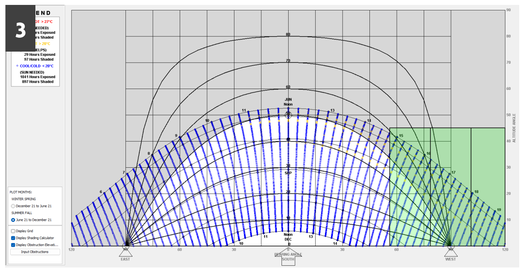


LATITUDE,LONGITUDE: **60.72 N,135.07 W**
KOPPEN CLIMATE CLASSIFICATION: **Sub Artic**
AVERAGE ANNUAL TEMPERATURE: **0 Degrees**
HIGHEST TEMPERATURE : **27 Degrees**
LOWEST TEMPERATURE: **-45 Degrees**
RELATIVE HUMIDITY: **65%**
LOW ANNUAL PRECIPITATION: **160.9 mm**
ANNUAL SNOW FALL: **1,418 mm**









CLIMATE IMPLICATIONS ON DESIGN

Figure 1: Temperature Range Chart
Indicates the extreme cold temperature range of the site with annual average temperature of 0 degrees which is far below the comfort level.

Figure 2: Dry bulb and humidity
Relative humidity recorded each month is above the comfort at 65%. This requires dehumidification of spaces to reach the comfort level.

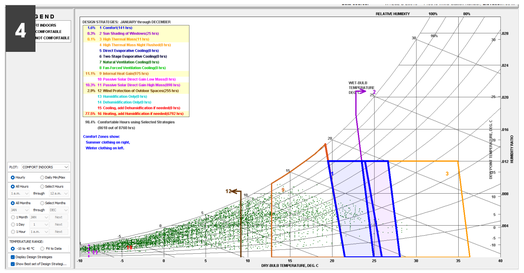


Figure 3: Sun Shading chart

- South facing glazing to maximize winter solar heat gain
- South glazing require sun shading at an angle of 50 degrees to shade the summer sun.
- Shaded green indicates deciduous trees located at north, north-west and north-east windows for shade

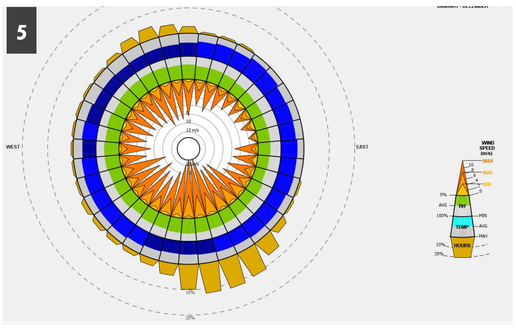


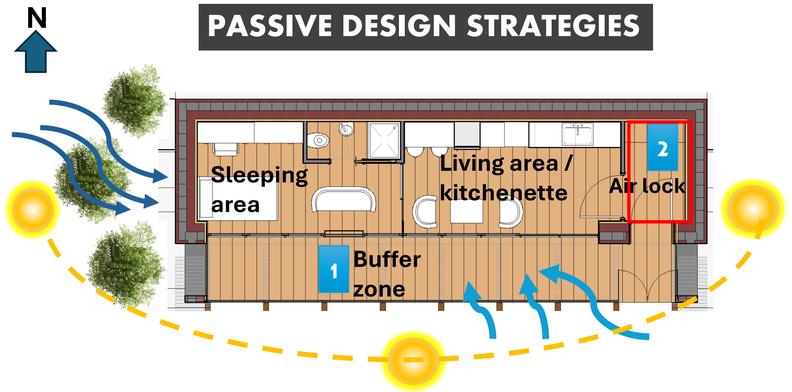
Figure 4: Psychrometric Chart
The following will lead to optimal comfort for users :

- Sun shading of windows
- High Thermal mass
- Internal heat gain
- Passive solar gain using high mass
- Wind protection of outdoor spaces

Figure 5: Wind Wheel
Direction of wind of this site is mainly from south-west and north-west. Annual wind speed is 5m/s.

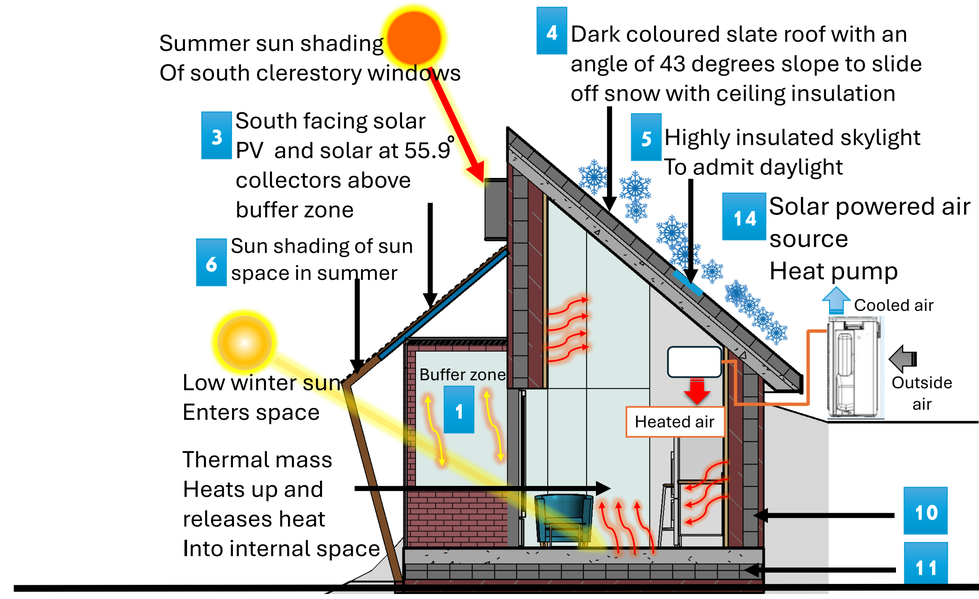
- Providing optimum comfort in this climate will require a combination of passive strategies and mechanical ventilation, heating and cooling systems; Energy recovery ventilators, Air source heat pump and solar hot water systems

PASSIVE DESIGN STRATEGIES



FLOOR PLAN OF SHIPPING CONTAINER

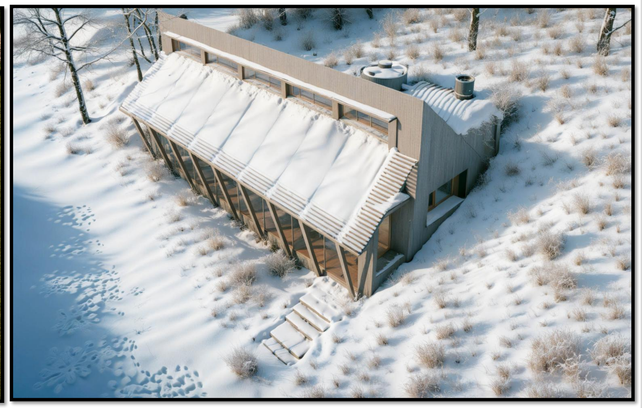
Orientation: Longest side faces south for solar passive heat gain
Building Form: Very compact building to keep heat within



CROSS SECTION OF SHIPPING CONTAINER



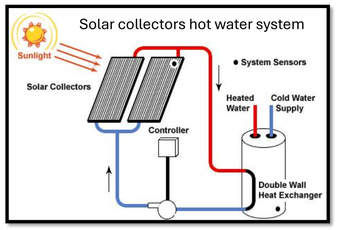
SOUTH –EAST VIEW IN SUMMER

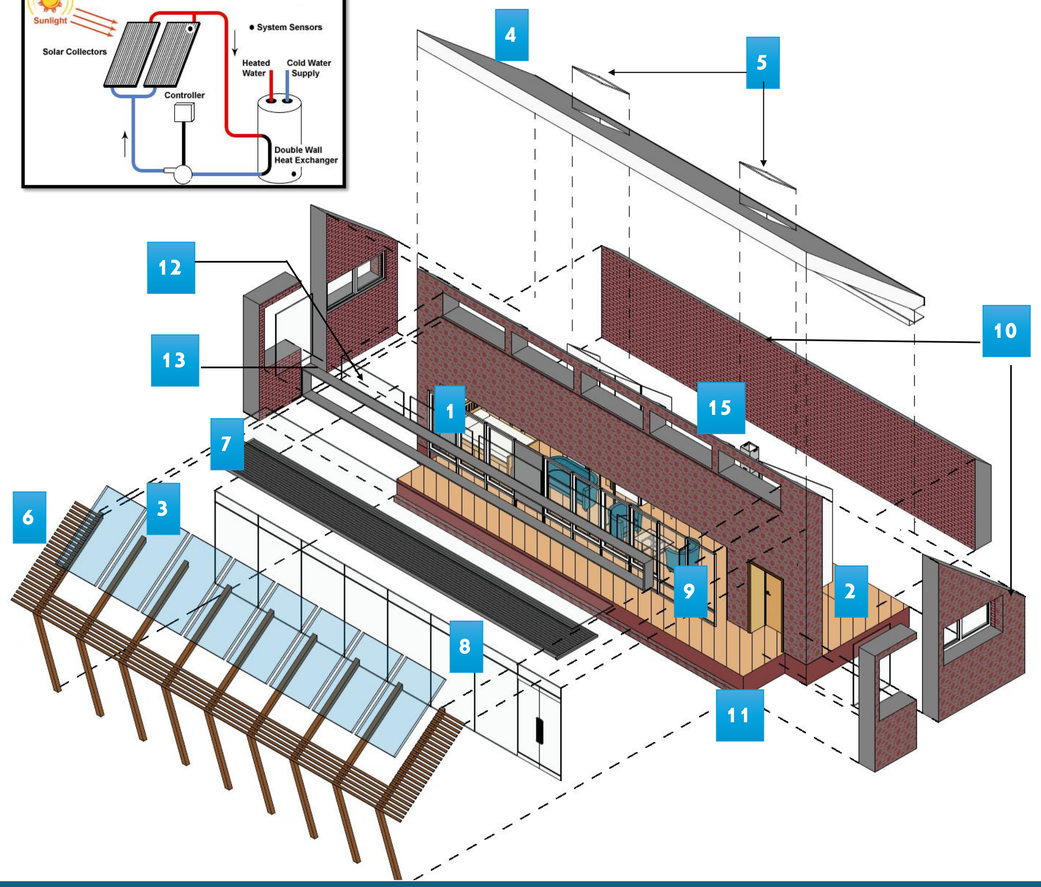


SOUTH –EAST VIEW IN WINTER

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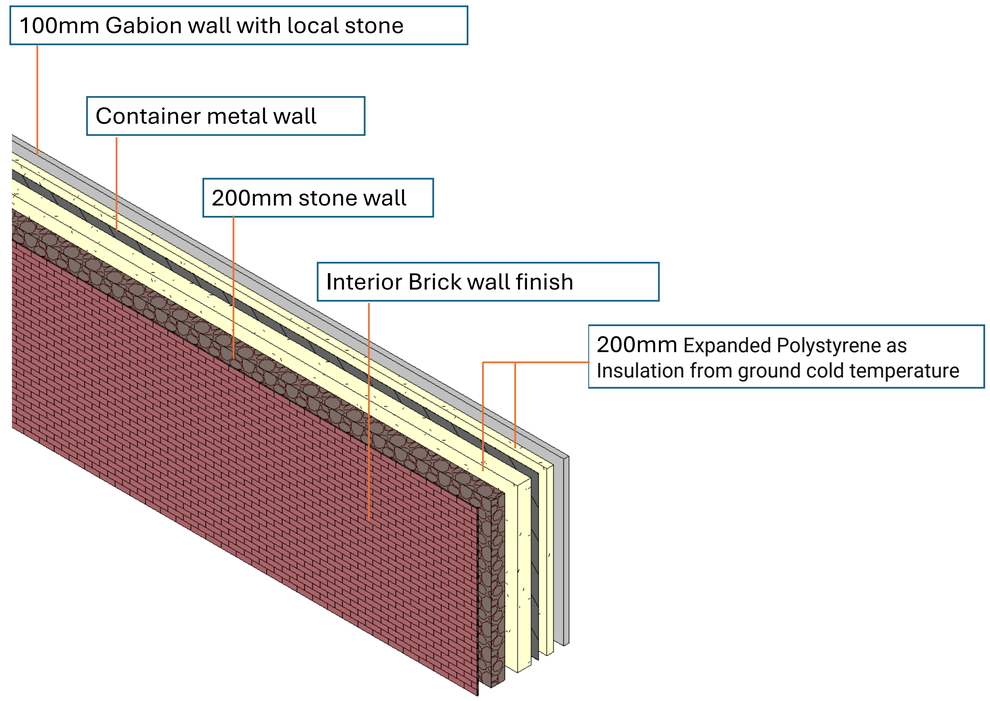
CLIMATE SENSITIVE SHIPPING CONTAINER - Cold Climate



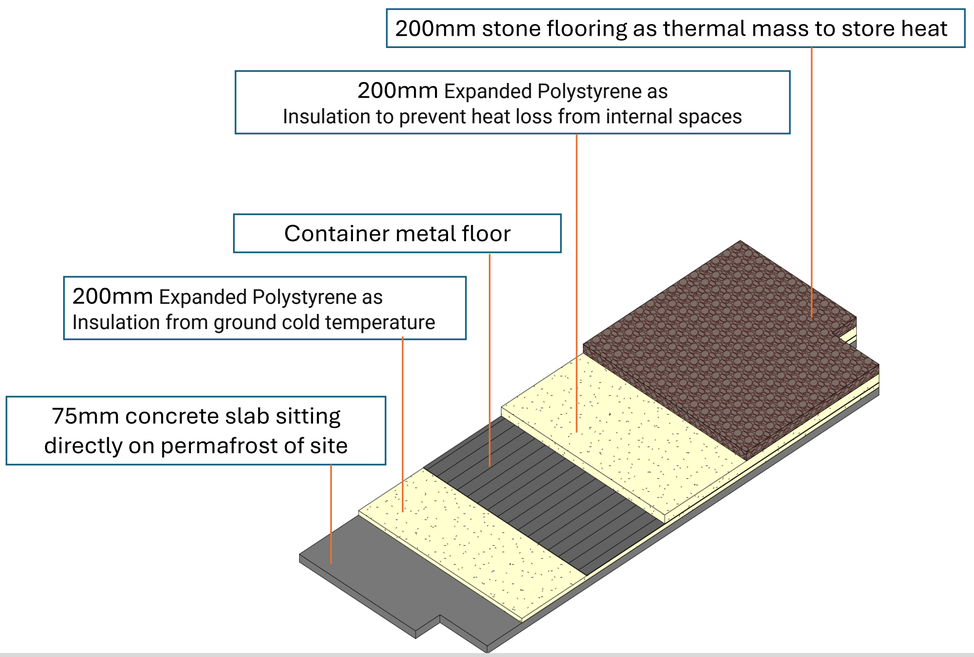


LEGEND	
1	Buffer zone providing thermal protection of the interior space
2	Air lock to minimize infiltration of cold air
3	Solar PV panels installed above buffer zone to generate power and shade the buffer zone during summer
4	Dark coloured insulated slate roof at an angle of 43 degrees slope to slide off snow. U value = 0.01
5	Highly insulated skylight to admit daylight
6	Solar shading 1 to reduce sun during summer
7	Solar shading 2 to reduce sun penetration during summer
8	Dynamic glazing wrapping buffer zone to admit daylight and winter sun
9	Argon filled tripled glazed window
10	Thermal mass (stone wall with insulation)
11	Insulated concrete and stone floor slab
12	South clerestory triple pane glass filled with argon
13	Horizontal shading hood over south clerestory glass
14	Solar powered air source heat pump
15	Extra heat gain from occupants and equipment indoors
16	Water tank 1& 2 wrapped tightly in insulation

NORTH WALL DETAIL:
U value = 0.0125



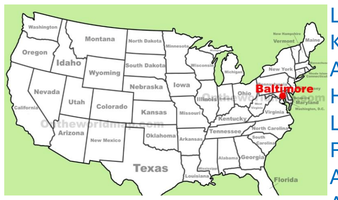
FLOOR SLAB DETAIL:
U value = 0.01



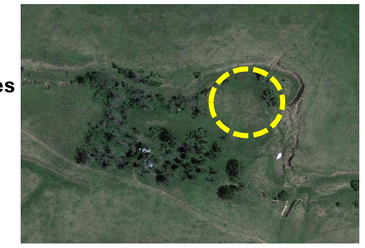
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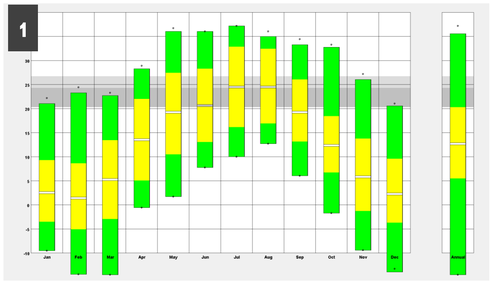
CLIMATE SENSITIVE SHIPPING CONTAINER - Temperate Climate

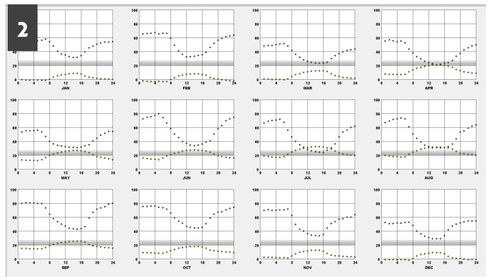
LOCATION : Clayton, New Mexico, UNITED STATES.

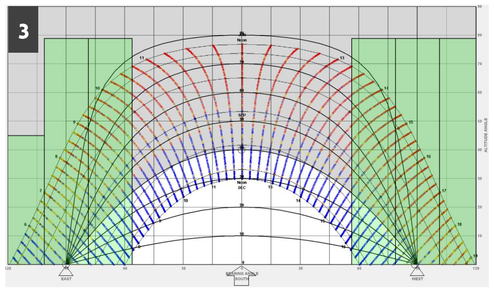


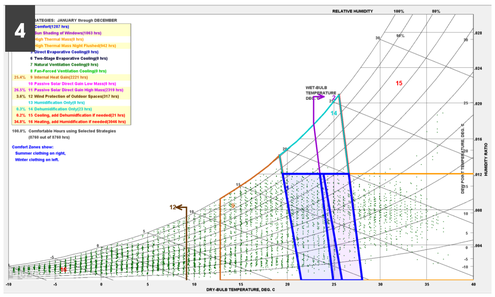
LATITUDE,LONGITUDE: **36.45 N, 103.15 W**
KOPPEN CLASSIFICATION: **Cold semi-arid**
AVERAGE ANNUAL TEMPERATURE:**13 Degrees**
HIGHEST TEMPERATURE : **36 Degrees**
LOWEST TEMPERATURE: **-14 Degrees**
RELATIVE HUMIDITY: **57%**
ANNUAL PRECIPITATION: **408 mm**
ANNUAL SNOW FALL: **685 mm**

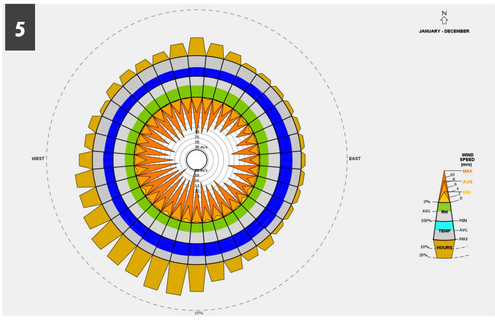












CLIMATE IMPLICATIONS ON DESIGN

Figure 1: Temperature Range Chart
A cold climate with warm temperatures and annual average temperature of 13 degrees.

Figure 2: Dry bulb and humidity
The average relative humidity of 57% is above the comfort level. March, April and July present humidity at the comfort level. Requires dehumidification to reach the comfort level.

Figure 3: Sun Shading chart
South facing glazing to maximize winter solar heat gain
South glazing require horizontal sun shading at an angle of 40 degrees and vertical shading from East and West at 60 degrees.
The area shaded green indicates deciduous trees located at N, NW, SW, E, NE and SE for shade.

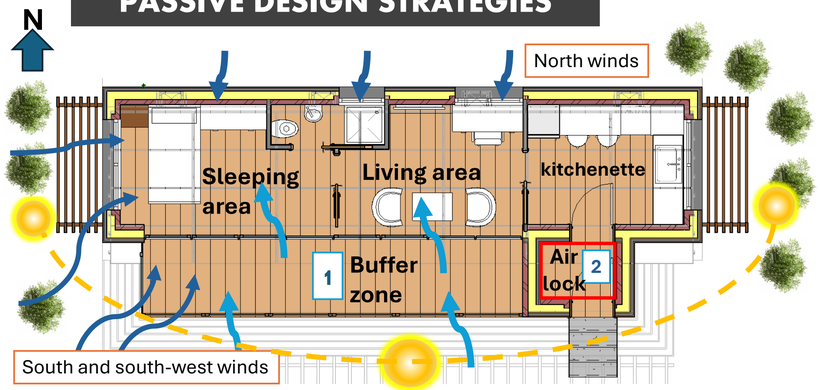
Figure 4: Psychrometric Chart
The following strategies will lead to optimal comfort:

- Sun shading of windows on east and west facades
- Thermal mass and passive solar gain
- Night flushing
- Wind protection with trees
- Dehumidification required
- 34.8% heating period in a year

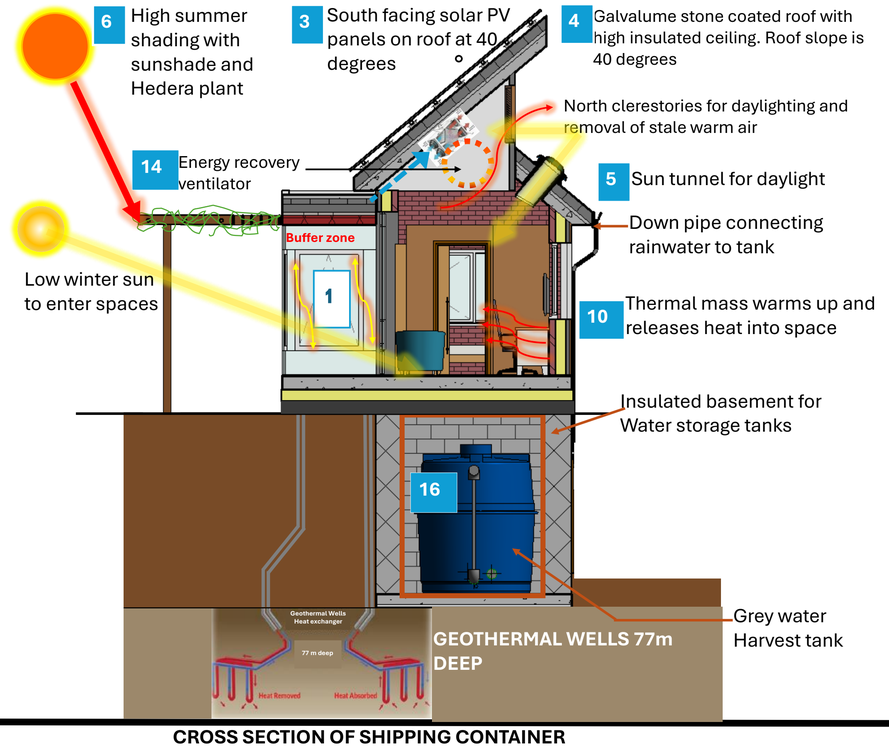
Figure 5: Wind Wheel
Direction of wind of this site is mainly from south-west and north. Annual wind speed is 6m/s.

- **Passive strategies are critical**
- **Natural ventilation is beneficial in this climate for half of the year.**
- **Active strategies required especially for heating**

PASSIVE DESIGN STRATEGIES



Orientation: Longest side faces south for solar passive heat gain
Building Form: Compact building to keep heat within



CROSS SECTION OF SHIPPING CONTAINER

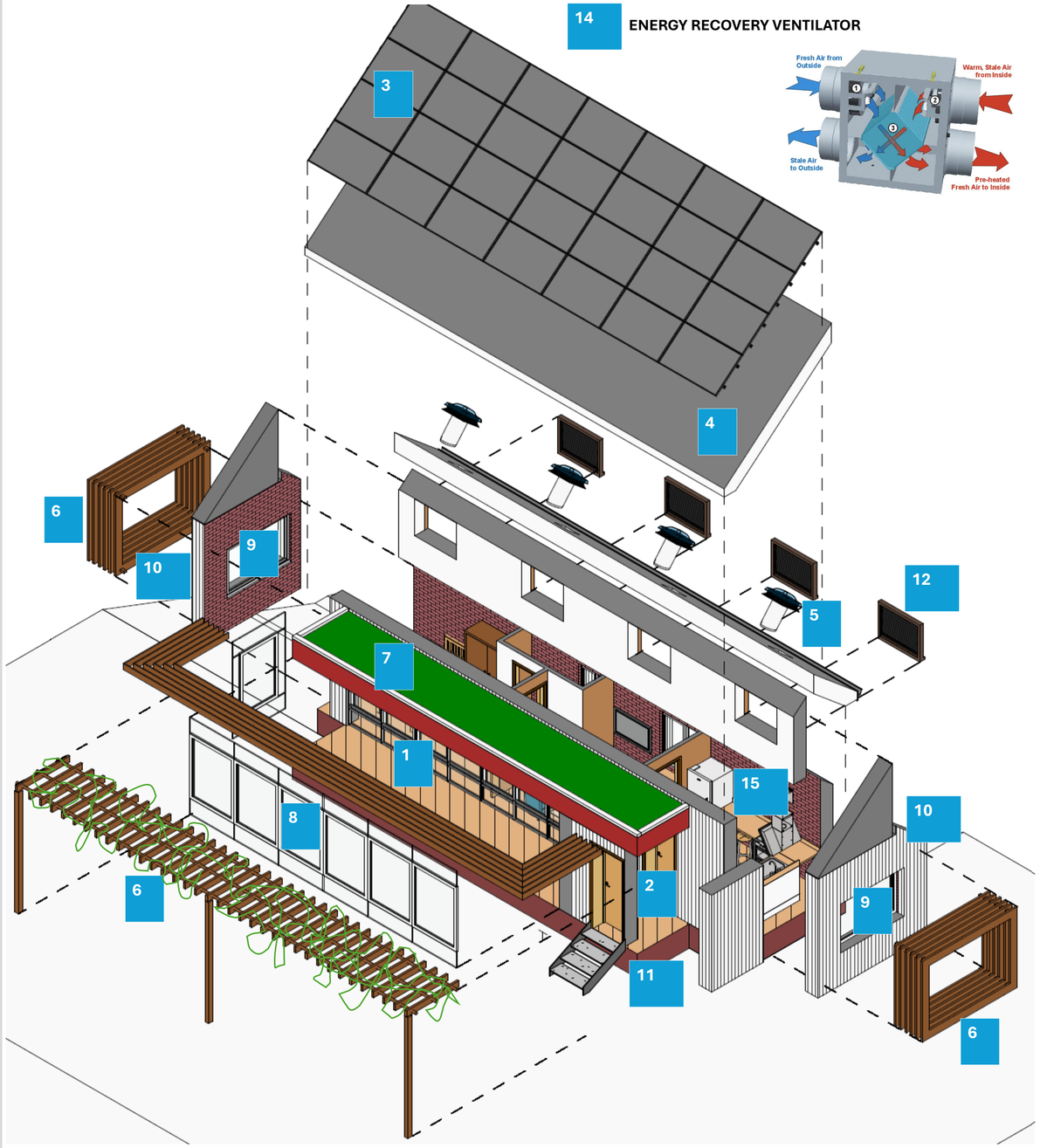

SOUTH –WEST VIEW


NORTH –WEST VIEW

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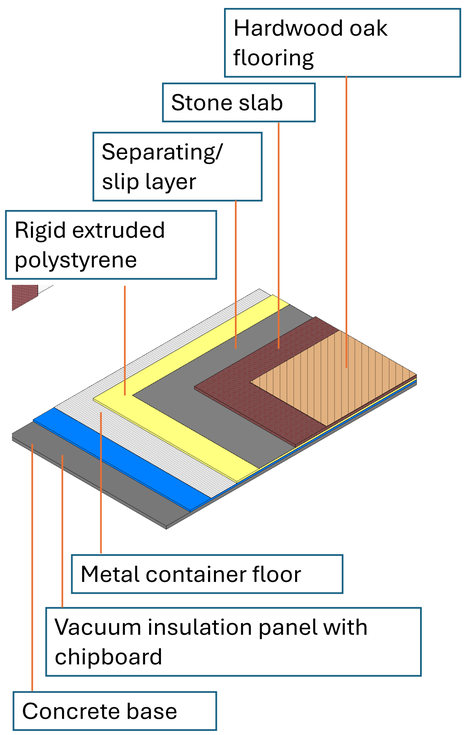
CLIMATE SENSITIVE SHIPPING CONTAINER - Temperate Climate

EXPLODED VIEW OF CONTAINER

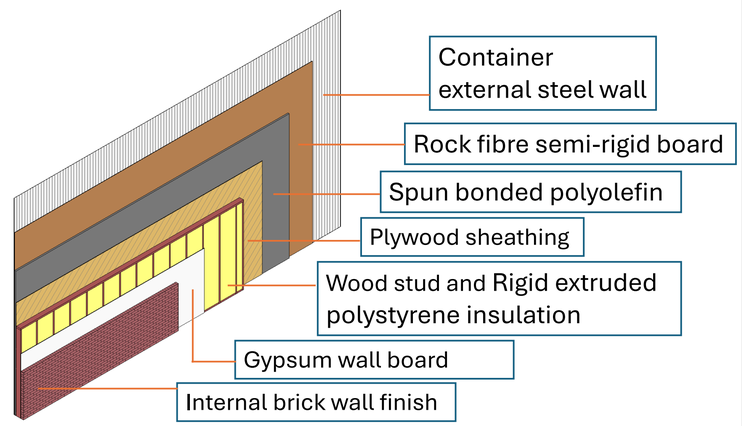


LEGEND	
1	Buffer zone providing thermal protection of the interior space
2	Air lock to minimize infiltration of cold air
3	Solar PV panels installed above buffer zone to generate power and shade the buffer zone during summer
4	Galvalume stone coated roof with high insulated ceiling. Roof slope is 40 degrees. **U-value:0.02**
5	Solar tunnels to admit daylight
6	Solar shading to reduce sun penetration during summer
7	Green roof with native Bouteloua curtipendula grass
8	Dynamic glazing wrapping buffer zone to admit daylight and winter sun
9	Argon filled tripled glazed window
10	Insulated wall with thermal mass *(wall detail below)*
11	Concrete and stone floor slab finished with oak
12	North clerestory for daylight and removal of stale air
13	Horizontal shading hood over south clerestory glass
14	Energy Recovery ventilator
15	Extra heat gain from occupants and equipment indoors
16	Water tanks to harvest rainwater, grey water and treated into clean tank to supply container

FLOOR DETAIL



WALL DETAIL: U value = 0.0285



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CLIMATE SENSITIVE SHIPPING CONTAINER – Warm Humid Climate

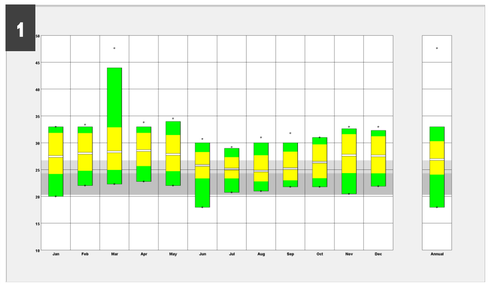
LOCATION : Accra, Ghana, Africa.



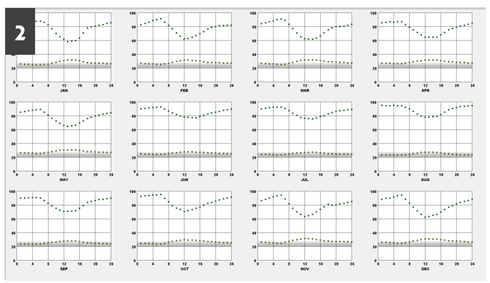
LATITUDE,LONGITUDE: **5.6 N, 0.17 W**
KOPPEN CLASSIFICATION: **Tropical Monsoon**
AVERAGE ANNUAL TEMPERATURE:**27 Degrees**
HIGHEST TEMPERATURE : **44 Degrees**
LOWEST TEMPERATURE: **17 Degrees**
RELATIVE HUMIDITY: **83%**
ANNUAL PRECIPITATION: **740 mm**
ANNUAL SNOW FALL: **NONE IN THIS CLIMATE**



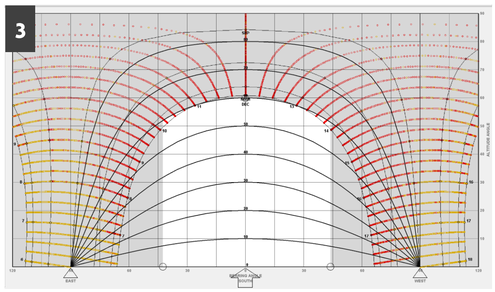
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2



3



CLIMATE IMPLICATIONS ON DESIGN

Figure 1: Temperature Range Chart
Extremely warm temperature location with annual average temperature of 27 degrees.

Figure 2: Dry bub and humidity
The average relative humidity of 83% is above the comfort level. Requires dehumidification and cooling during summer to reach the comfort level.

4

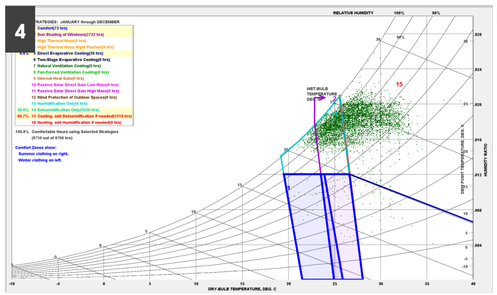


Figure 3: Sun Shading chart

- Shading windows on all sides required.
- Possibility of eliminating west windows to prevent intense heat gain.
- Vegetation around building to minimize heat gain
- Shading semi-open and open spaces
- Low-E glass in wooden frames to reduce heat gain

Figure 4: Psychrometric Chart
The following will lead to optimal comfort :

- Sun shading of all windows
- Light weight walls and light-coloured building materials to reduce heat gain.
- -Well ventilated pitched roof
- Raise building above ground to reduce dampness
- Dehumidification required
- Long narrow floor plans to maximize cross ventilation

5

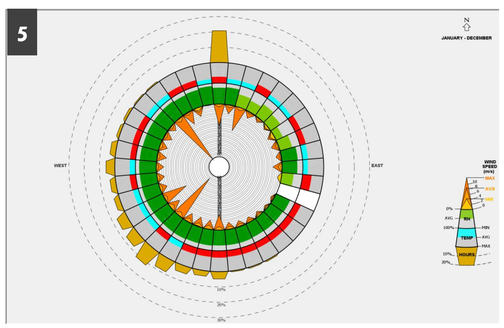
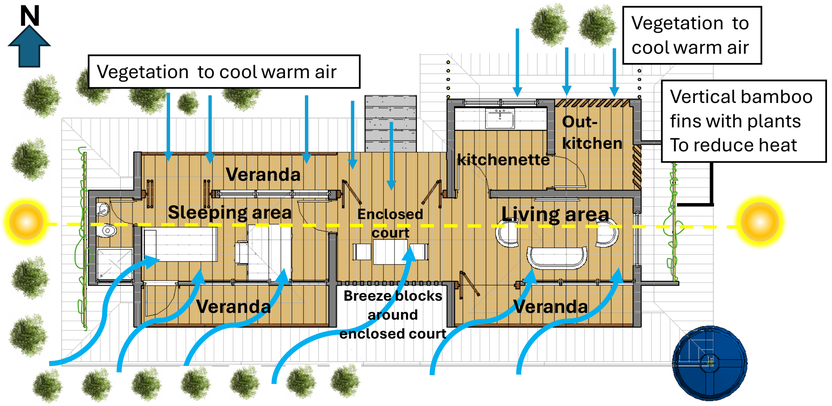


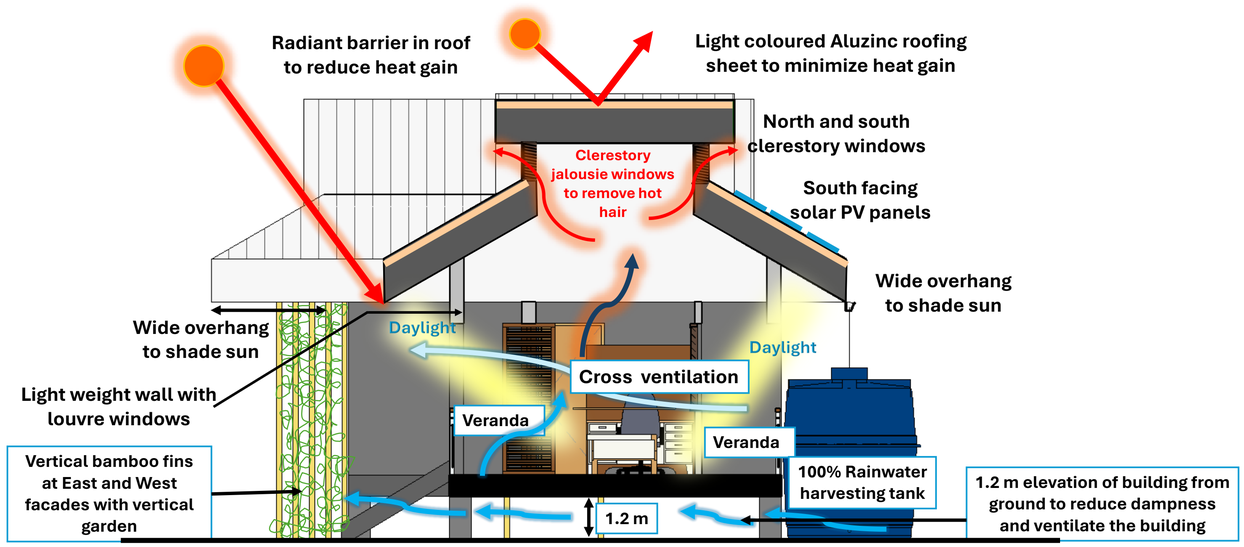
Figure 5: Wind Wheel
Direction of wind is mainly from south-west and north. Annual wind speed is 6m/s.
Orient windows towards prevailing winds.

- **Passive strategies are excellent to achieve comfort**
- **100% natural ventilation and lighting possible throughout the year.**
- **Very minimal inclusion of active systems**

PASSIVE DESIGN STRATEGIES



Building Form: Long narrow plan with operable windows for ventilation
Orientation: Longest side faces north & south for optimum ventilation



CROSS SECTION OF SHIPPING CONTAINER



North – East View

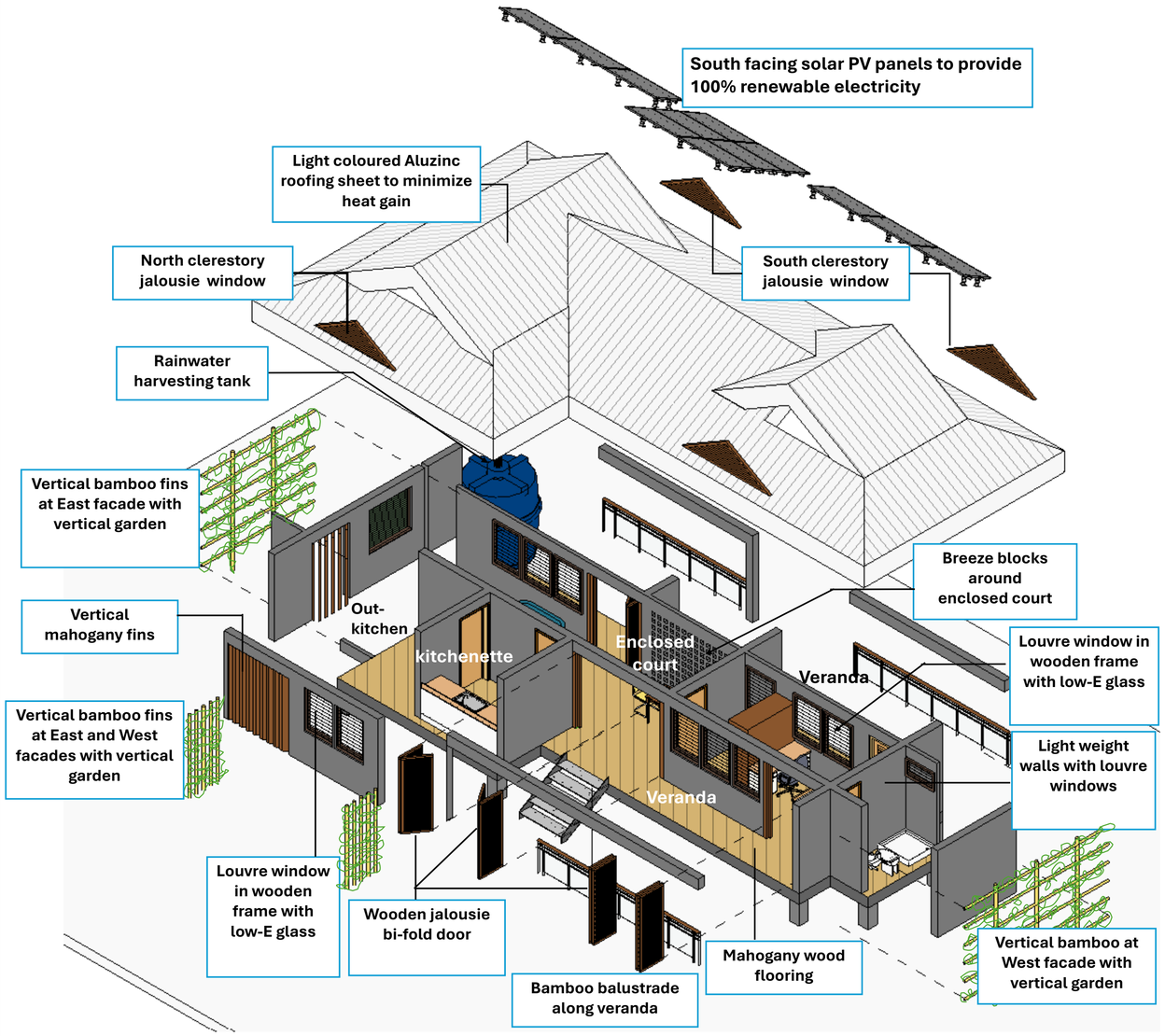


North – West View

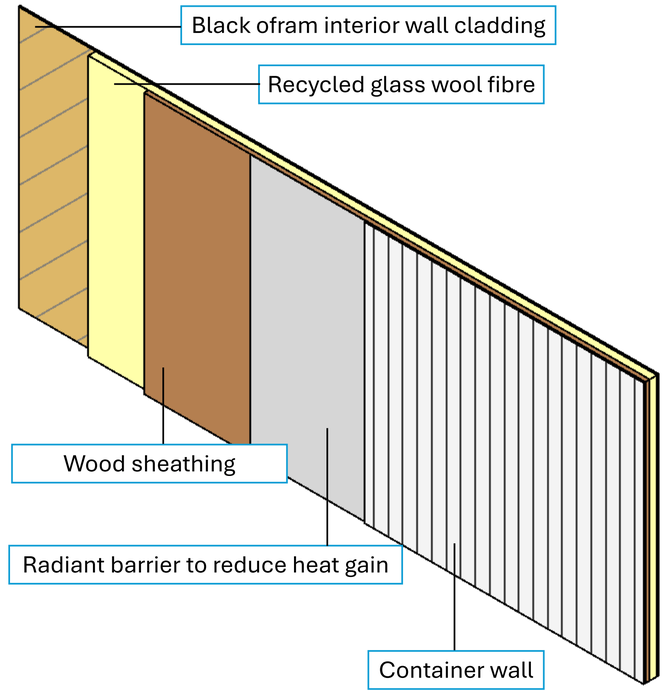
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CLIMATE SENSITIVE SHIPPING CONTAINER – Warm Humid Climate

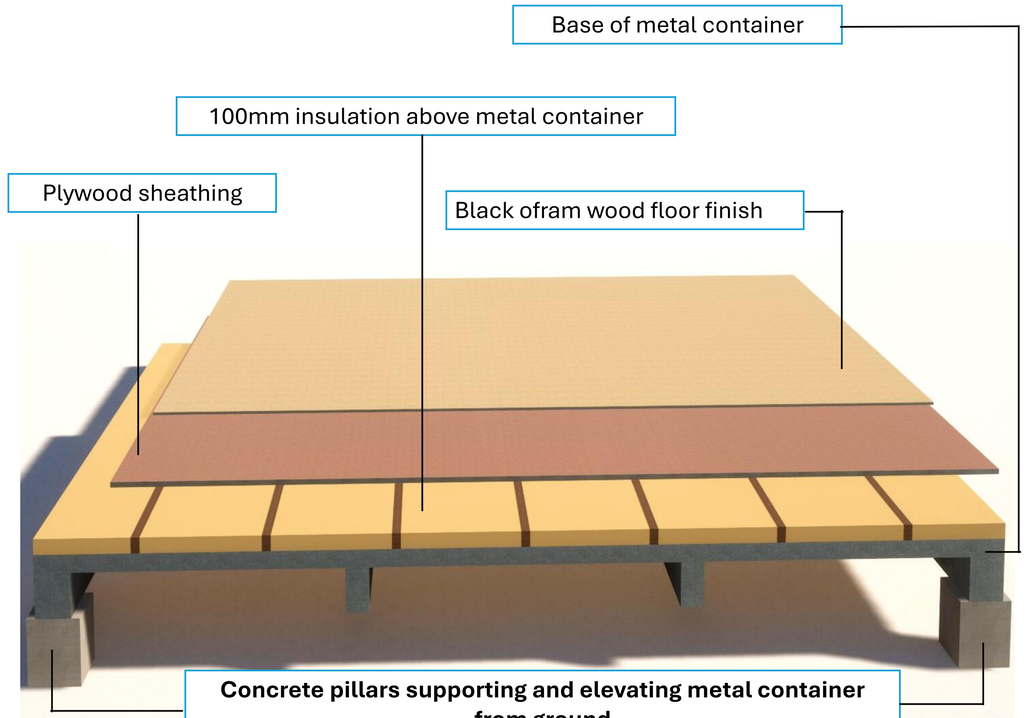
EXPLODED VIEW OF CONTAINER



WALL DETAIL



SLAB DETAIL



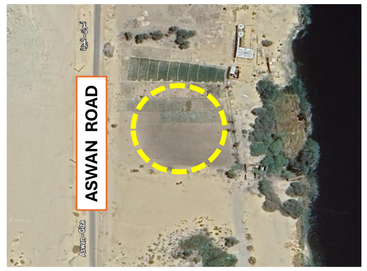
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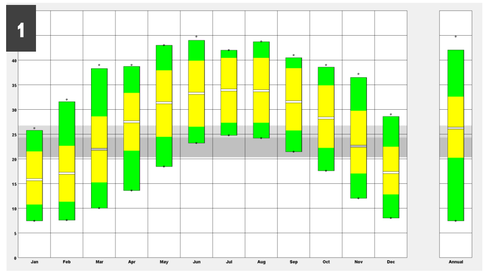
CLIMATE SENSITIVE SHIPPING CONTAINER – Hot Dry Climate

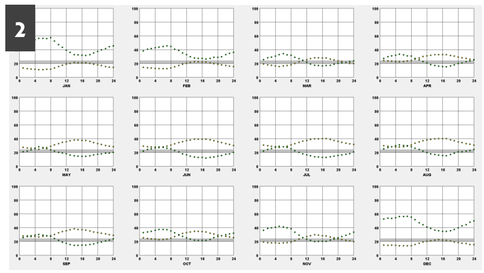
LOCATION : Aswan, EGYPT, Africa.

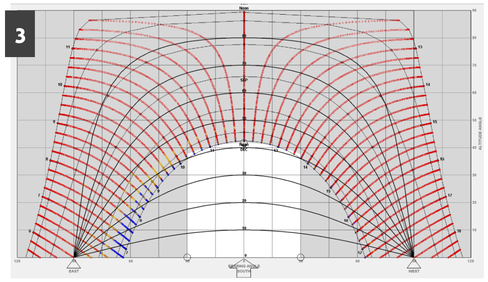


LATITUDE,LONGITUDE: **23.97 N, 32.78 E**
KOPPEN CLASSIFICATION: **Arid desert hot**
AVERAGE ANNUAL TEMPERATURE:**27 Degrees**
HIGHEST TEMPERATURE : **45 Degrees Celsius**
LOWEST TEMPERATURE: **7 Degrees**
RELATIVE HUMIDITY: **26%**
ANNUAL PRECIPITATION: **2 mm**
ANNUAL SNOW FALL: **NONE**









CLIMATE IMPLICATIONS ON DESIGN

Figure 1: Temperature Range Chart
This presents an extremely hot and dry temperature location with annual average temperature of 27 degrees which is above the comfort level. Cooling is urgent.

Figure 2: Dry bulb and humidity
The average relative humidity of 26% indicates a very dry climate requiring humidification to reach the comfort level.

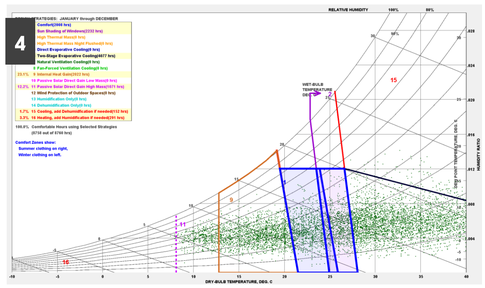


Figure 3: Sun Shading chart
- Small recessed shaded windows required.
- Possibility of eliminating west windows to prevent intense heat gain.
- Vegetation around building to minimize heat gain
- Double pane high performance glazing to eliminate heat into spaces.

Figure 4: Psychrometric Chart
Focusing on the following will lead to optimal comfort:
- Sun shading of all windows
- Two stage evaporative cooling
- Internal heat gain at cold nights
- High mass for direct heat gain at night
- Cooling required
- Heating at cold nights

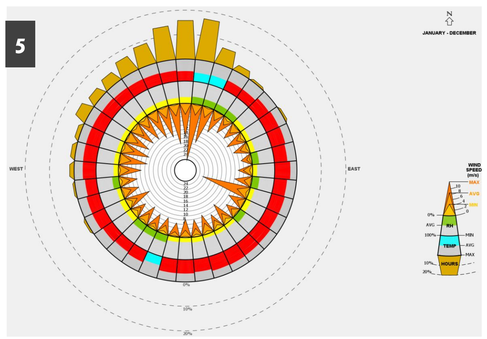
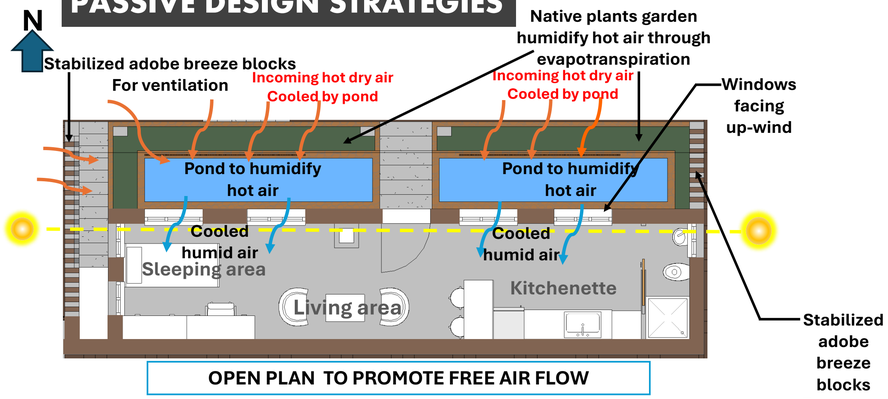
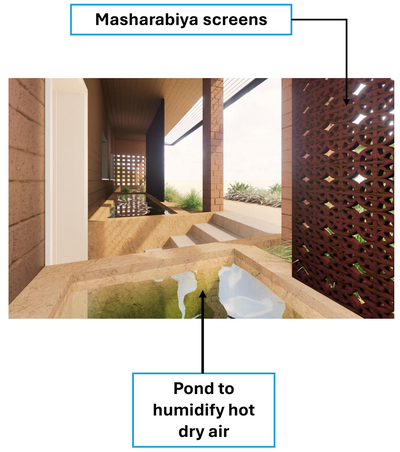


Figure 5: Wind Wheel
Direction of wind of this site is mainly from north-west and north. Annual wind speed is 4m/s.
Orient windows towards prevailing winds.

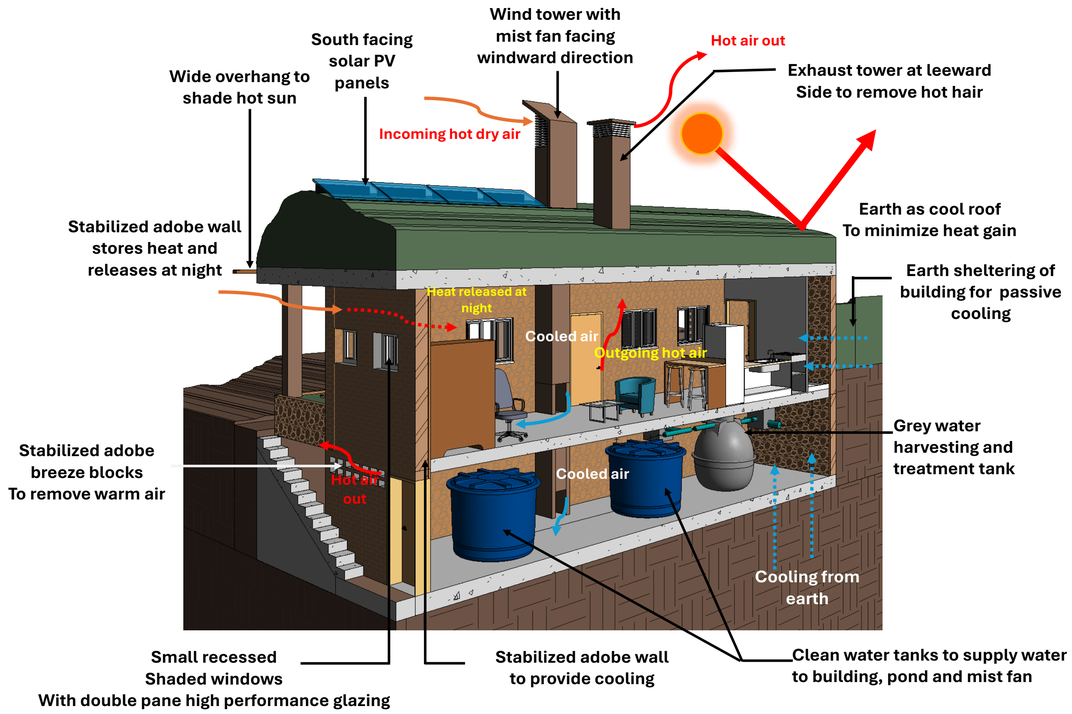
- **Passive strategies are excellent to achieve comfort**
- **Achieving natural ventilation**
- **Inclusion of minimal active systems for cooling**

PASSIVE DESIGN STRATEGIES





Orientation: Building faces North and North-East for optimum ventilation
Building Form: Compact earth sheltered building with passive strategies





North view showing vegetation cover over the building

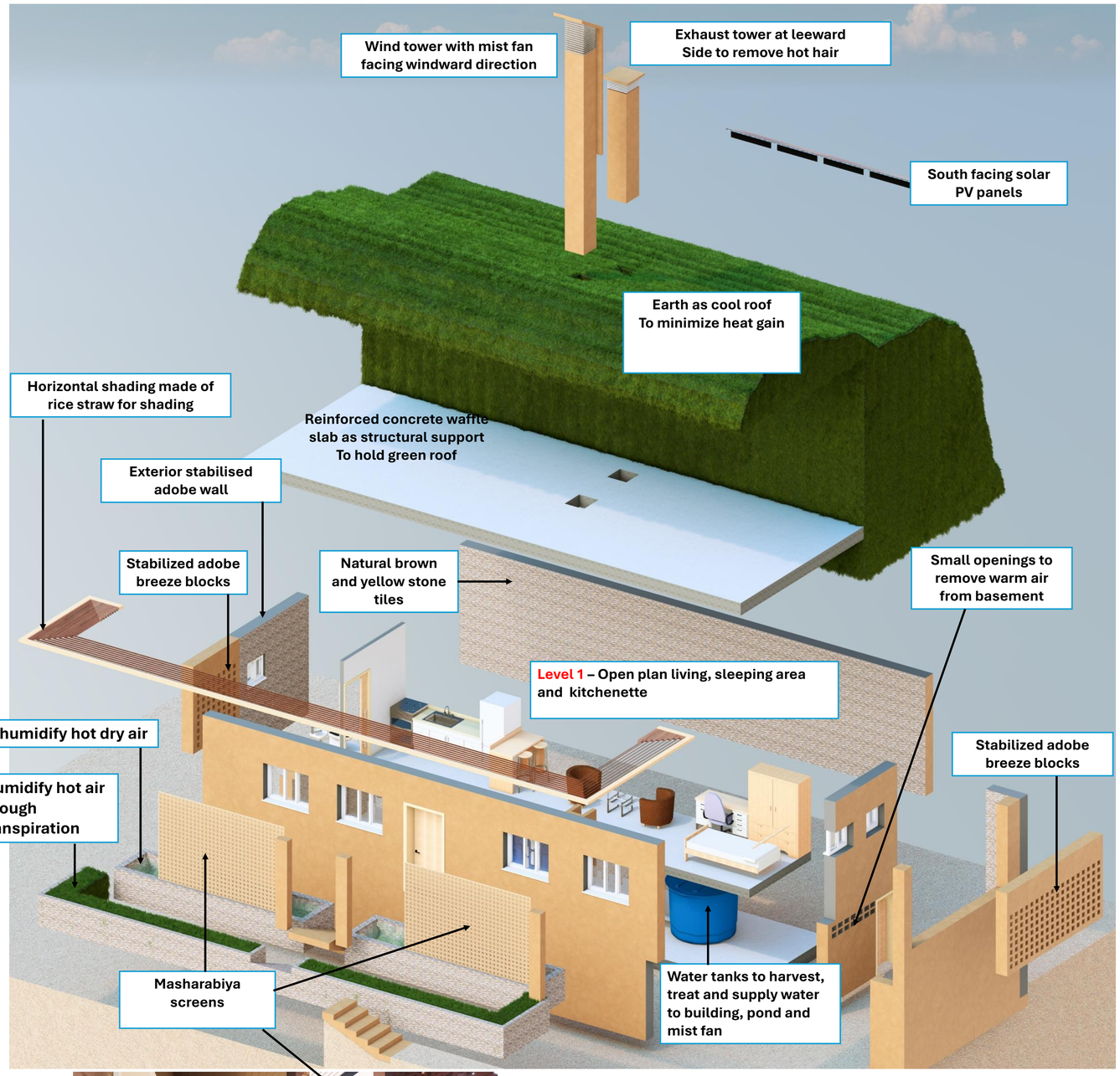


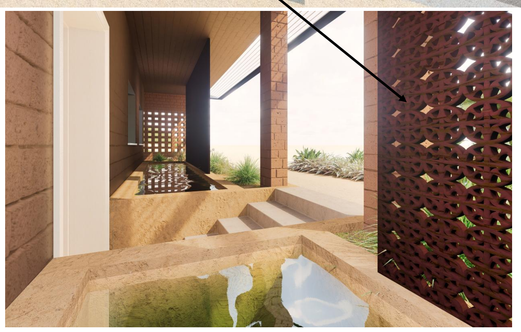
North-West view

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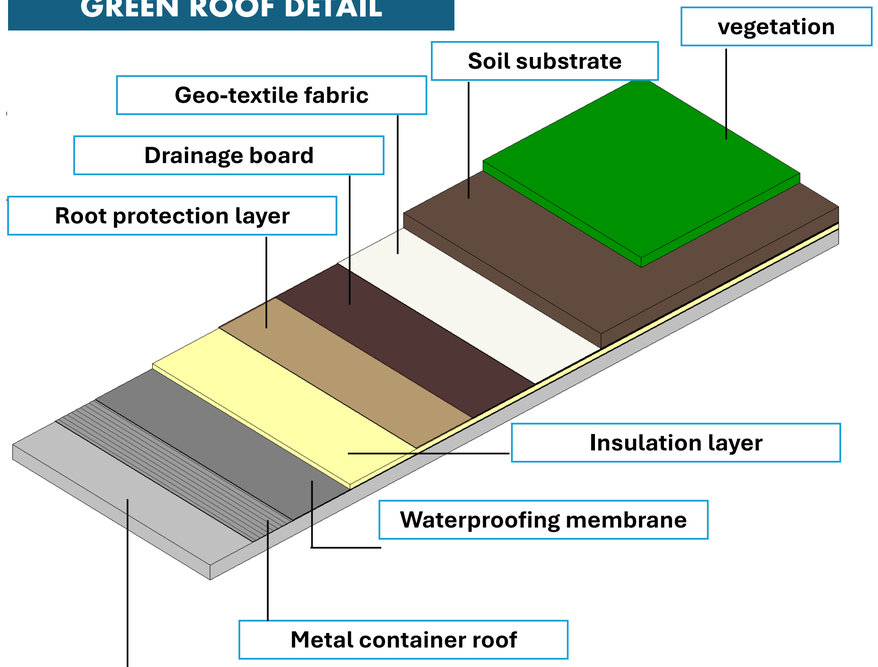
CLIMATE SENSITIVE SHIPPING CONTAINER – Hot Dry Climate

EXPLODED VIEW OF CONTAINER

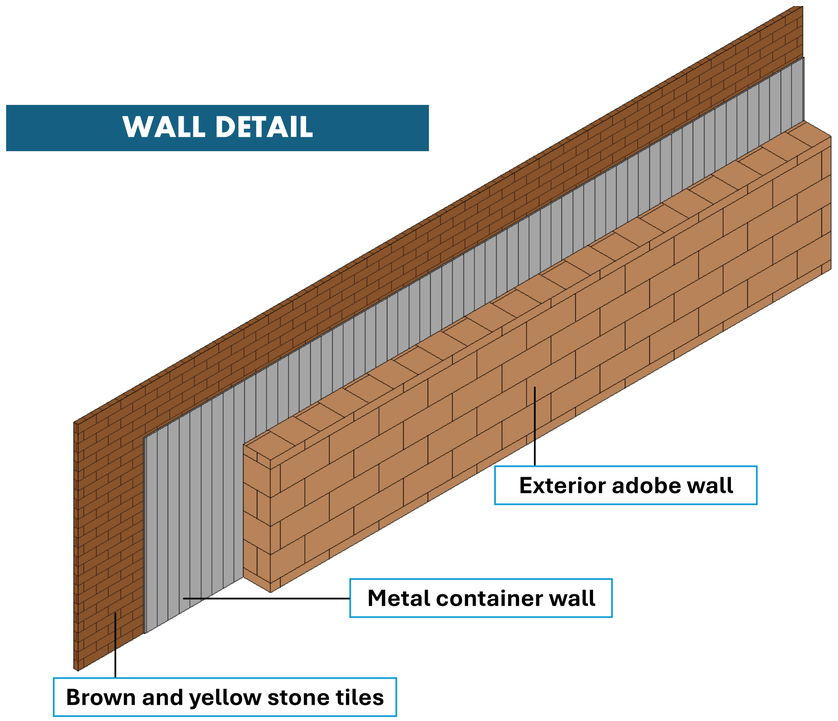




GREEN ROOF DETAIL



WALL DETAIL



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CLIMATE SENSITIVE SHIPPING CONTAINER

BIBLOGRAPHY

Elrayies, G.M. (2017) 'Thermal performance assessment of shipping container architecture in hot and humid climates', *International Journal on Advanced Science, Engineering and Information Technology*, 7(4), pp. 1114–1126. Available at : https://doi.org/10.18517/ijaseit.7.4.2235

Jonas, T. (2015) 'Storage, study, sleeping: Container convert', *ReNew: Technology for a Sustainable Future*, (133), pp. 48–51. Available at: https://www.jstor.org/stable/renetechsustfutu.133.48

Kernohan, K. (2013) 'Self-contained living: Shipping container reuse', *ReNew: Technology for a Sustainable Future*, (122), pp. 36–39. Available at: https://www.jstor.org/stable/renetechsustfutu.122.36

Kristiansen, A.B., Zhao, B.Y., Ma, T. and Wang, R.Z. (2021) 'The viability of solar photovoltaic powered off-grid Zero Energy Buildings based on a container home', *Journal of Cleaner Production*, 286, p. 125312. Available at: doi: 10.1016/j.jclepro.2020.125312

Pereira-de-Oliveira, L.A., Bernardo, L.F.A. and Marques, A.R.A. (2022) 'Architectural building design with refurbished shipping containers: A typological and modular approach', *Journal of Engineering Research*, 10(1A), pp. 1–20. Available at: doi: 10.36909/jer.10853

Residential Shipping Container Primer RSCP (n.d.) *Shipping container structural components and terminology*. Available at: Residential Shipping Container Primer (RSCP™) – Building with Shipping Containers: The Practical Handbook

Zhang, G., Setunge, S. and van Elmpt, S. (2014) 'Using shipping containers to provide temporary housing in post-disaster recovery: Social case studies', *Procedia Economics and Finance*, 18, pp. 618–625. Available at: 10.1016/S2212-5671(14)00983-6

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